Mohammed Abdalazeem

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EDUCATION

University of Massachusetts Amherst — *Jan 2021 – Expected Fall 2026*
PhD in Civil Engineering (Transportation Systems) — GPA: 3.92 — *Amherst, MA*

- Dissertation: *Machine Learning and Optimization for Schematic Bicycle Map Design and Cycling Behavior*
- Advisor: Dr. Jimi Oke

University of Massachusetts Amherst — *Sep 2023 – Dec 2025*
MS in Computer Science — GPA: 3.71 — *Amherst, MA*

University of Massachusetts Amherst — *Jan 2021 – May 2024*
MS in Civil Engineering — GPA: 3.92 — *Amherst, MA*

University of Khartoum — *Sep 2013 – Nov 2018*
BSc in Civil Engineering, First Class Honors — *Khartoum, Sudan*

AWARDS & HONORS

- CEE Departmental Fellowship, University of Massachusetts Amherst (2023)
- Graduate School Travel Awards, University of Massachusetts Amherst (2022–2024)
- First Class Honors (Summa Cum Laude Equivalent), BSc Civil Engineering, University of Khartoum (2018)
- Best Project Award, Department of Civil Engineering, University of Khartoum (2018)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — *Jan 2021 – Present*
University of Massachusetts Amherst — *Amherst, MA*

- Research on micromobility map design, road safety, and transit demand modeling.

Teaching Assistant — *Fall 2023 – Fall 2025*
University of Massachusetts Amherst — *Amherst, MA*

- TA for probability & statistics and transportation systems courses.

Instructor, Department of Civil Engineering — *Nov 2018 – Dec 2020*
University of Khartoum — *Khartoum, Sudan*

- Taught and led tutorials for five undergraduate civil engineering courses.

PUBLICATIONS

Journal Articles (Published)

1. Ruger, A., Abdalazeem, M., Christofa, E., & Oke, J. (2026). Explainable Data-Driven Multi-Filter Framework for Bot Detection in Incentivized Online Surveys. *Data Science for Transportation*, 8, 21. DOI
2. Abdalazeem, M., Altayb, G., Christofa, E., & Oke, J. (2026). Maps are good but are simpler maps better? Insights on urban bicycling in the US. *Environmental Research Communications*, 8(2), 025004. DOI
3. Zhao, P., Mega, J., Abdalazeem, M., Arabi, M., Barchers, C. V., & Oke, J. (2026). Scenario discovery framework aids robust regional emissions mitigation planning. *Environmental Research Communications*, 8(3), 031008. DOI
4. Abdalazeem, M., & Oke, J. (2025). A Roadway Crash Typology of Census Tracts Enables Targeted Interventions via Interpretable Machine Learning. *Data Science for Transportation*, 7, 14. DOI
5. Abdalazeem, M., & Oke, J. (2024). Enhanced Seasonal Typology-Informed Transit Trip Chaining via Mobile Boarding and Survey Data. *Data Science for Transportation*, 6, 19. DOI
6. Abdalazeem, M., & Oke, J. (2023). Extracting Spatiotemporal Bus Passenger Trip Typologies from Noisy Mobile Ticketing Boarding Data. *Data Science for Transportation*, 5, 20. DOI

7. Abdalazeem, M., & Oke, J. (2023). Origin-destination inference in public transportation systems: A comprehensive review. *International Journal of Transportation Science and Technology*, 12(1), 315–328. DOI

Conference Papers

1. Abdalazeem, M., Mohammed, O., Ibrahim, M., & Osman, H. G. (2020). Khartoum sky trains network: A study on implementing caterpillar trains concept in Khartoum. *MATEC Web of Conferences*, 308, 02003. DOI

Under Review

1. Abdalazeem, M., Christofa, E., Barchers, C., & Oke, J. Toward Unified Design Rules for Schematic Bicycle Maps.
2. Abdalazeem, M., Ruger, A., Christofa, E., & Oke, J. Schematic maps promise to induce micromobility ridership: a Bayesian latent class ordered logistic approach.
3. Bohlke, N., Bell, H., Abdalazeem, M., Oke, J., & Christofa, E. Three-decade systematic review highlights enduring barriers to cycling and reveals rural–urban evidence gap.

In Preparation

1. Abdalazeem, M., Demers, J., Christofa, E., & Oke, J. A Unified Geospatial Data Pipeline and Detection Algorithm for Bicycle Routes from Heterogeneous Network Data.
2. Abdalazeem, M., Christofa, E., & Oke, J. Automated Schematization of Bicycle Route Networks via Mixed-Integer Optimization.
3. Abdalazeem, M., Oke, T., & Oke, J. A novel origin-destination-transfer model using mobile ticketing activations with seasonal passenger typology and spatial error correction.
4. Zhao, P., Weinman, A., Rost-Nasshan, V., Abdalazeem, M., Han, Z., Arabi, M., & Oke, J. Road Network Typology of US Metropolitan Areas Explains Congestion, Mobile Emissions, and Safety Outcomes.

CONFERENCE PRESENTATIONS

Lectern Presentations

1. Abdalazeem, M., Oke, T., & Oke, J. (2025). A Typology-Informed Origin-Destination-Transfer Model for a Bus Transit Network using Mobile Ticketing Data. *TRB Annual Meeting, Washington, D.C.*
2. Abdalazeem, M., & Oke, J. (2024). Enhancing Road Safety: A Data-Driven Spatial Typology of Crashes in New England. *INFORMS Annual Meeting, Seattle, WA.*
3. Abdalazeem, M., & Oke, J. (2023). Typology-Enhanced Origin-Destination-Transfer Inference from Noisy Mobile Boarding Observations. *INFORMS, Phoenix, AZ.*
4. Abdalazeem, M., & Oke, J. (2023). Spatiotemporal Trip Chaining Framework for Open Mobile Fare Collection Systems. *TRB Annual Meeting, Washington, D.C.*
5. Abdalazeem, M., & Oke, J. (2022). Spatio-Temporal Trip Pattern Typology Analysis for a Regional Bus Network. *INFORMS, Indianapolis, IN.*

Poster Presentations

1. Abdalazeem, M., Christofa, E., & Oke, J. (2025). Optimizing and Deploying Schematic Bicycle Maps with MILP and User Input. *INFORMS Annual Meeting, Atlanta, GA.*
2. Abdalazeem, M., & Oke, J. (2025). A Roadway Crash Typology of Census Tracts Enables Targeted Interventions via Interpretable Machine Learning. *NEUTC Symposium, Norwich University, Northfield, VT.*
3. Abdalazeem, M., & Oke, J. (2025). A Spatial Typology Analysis of Crash Characteristics across 2,480 Census Tracts. *TRB Annual Meeting, Washington, D.C.*
4. Abdalazeem, M., & Oke, J. (2022). Trip Pattern Typologies in The Pioneer Valley Bus Transit System. *TRB Annual Meeting, Washington, D.C.*
5. Abdalazeem, M., & Oke, J. (2022). Efficient Extraction of Spatiotemporal Bus Passenger Trip Patterns in the Pioneer Valley Bus Transit System. *MassDOT Transportation Innovation Conference, Worcester, MA.*

RESEARCH EXPERIENCE

Graduate Research Assistant — *Jan 2021 – Present*
University of Massachusetts Amherst — *Amherst, MA*

Schematic Bicycle Map Design and Micromobility Behavior

- Mapped the national landscape of schematic bicycle map adoption across 684 US urban counties; Bayesian mixture modeling with XGBoost and SHAP analysis revealed three county types of adoption.
- Designed and deployed an incentivized national survey on how cyclists and micromobility riders respond to map design: simplification level, geographic accuracy, infrastructure representation, and route legibility.
- Modeled stated intentions to ride under different map designs with a Bayesian latent class ordered logit that separates riders into groups with different design sensitivities.
- Formulated a mixed-integer linear program that selects bicycle map routes from street and trail networks under safety, connectivity, and user-preference constraints.
- Built a bot detection pipeline for online surveys that combines platform signals, browser fingerprinting, timing checks, and an XGBoost validation model.
- *Funding: Armstrong Fund for Science, 2024–2026 (PI: J. Oke, co-PI: E. Christofa); NSF REU, 2023–2026.*
- *2 journal articles published, 3 under review, 2 in preparation.*

Spatial Crash Typology and Risk Prediction

- Classified 2,480 New England census tracts into crash types with UMAP and Gaussian mixture models, using one year of state-level crash records.
- Linked patterns in crash frequency and severity to roadway geometry, land use, and socioeconomic conditions across urban, suburban, and rural tracts.
- Trained XGBoost models with SHAP interpretation to predict crash risk within each type and show which factors drive risk where, informing where safety interventions should go.
- *Funding: New England University Transportation Center (NEUTC), 2024–2025 (PI: J. Oke).*
- *1 journal article published.*

Transit OD Inference from Partial Mobile Ticketing Data

- Built a framework that infers full origin-destination-transfer matrices from boarding-only mobile ticketing data on a regional bus network with 40+ routes.
- Clustered 40,000+ weekly riders into passenger types using hierarchical clustering with dynamic time warping.
- Added a gradient boosting spatial error correction step that cut MAE by 70% and SMAPE by 85% relative to the baseline trip chaining.
- *Funding: Pioneer Valley Transit Authority (PVTA), 2021–2023 (PI: J. Oke).*
- *3 journal articles published, 1 in preparation.*

TEACHING EXPERIENCE

Teaching Assistant — CE-ENGIN 260: Probability & Statistics for Civil Engineers — *Fall 2024, Fall 2025*
University of Massachusetts Amherst — *Amherst, MA*

- Led lab sections and weekly office hours covering probability, statistical inference, and experimental design.
- Supported roughly 160 students per semester; graded exams and wrote solution guides and grading rubrics.

Teaching Assistant — CE-ENGIN 310: Transportation Systems — *Fall 2023*
University of Massachusetts Amherst — *Amherst, MA*

- Supported students in transportation operations, network planning, and geometric design; prepared problem sets on traffic flow, capacity analysis, and signal control.

Teaching Assistant — Multiple Civil Engineering Courses — *Nov 2018 – Dec 2020*
University of Khartoum — *Khartoum, Sudan*

- Taught and led tutorials for undergraduate courses:
  - Environmental Engineering
  - Highway Engineering

  - Strength of Materials
  - Reinforced Concrete Design
  - Theory of Structures
- Supervised lab sessions on water quality analysis and material testing.

MENTORSHIP

Research Mentor, NSF REU Program *Summers 2023–2026*
University of Massachusetts Amherst *Amherst, MA*

- **Julianna Demers** (Rensselaer Polytechnic Institute), Summer 2026: Unified geospatial data pipeline for schematic bicycle map generation *(1 paper in preparation)*.
- **Andrew Ruger** (Grinnell College), Summer 2025: Survey data quality and micromobility mapping; bot detection in incentivized online surveys *(1 published paper, DOI; 1 paper under review)*.
- **Geehan Altayb** (Howard University), Summer 2024: Bicycle map features and their influence on cycling behavior *(1 published paper, DOI)*.
- **Vivian Rost-Nasshan** (Rensselaer Polytechnic Institute), Summer 2023: Vehicular delay patterns using network topology and demographic data *(1 paper in preparation)*.

Research Mentor *Spring 2026*
University of Massachusetts Amherst *Amherst, MA*

- **Henry Bell** (UMass Amherst), Spring 2026: Comprehensive literature review on barriers to cycling *(1 paper under review).*

PROFESSIONAL SERVICE

Manuscript Reviewing

- *Discover Public Health* (Springer Nature), 2026
- *Journal of Transportation Engineering, Part A: Systems* (ASCE), 2026
- *npj Environmental Social Sciences* (Springer Nature), 2026
- *Scientific Reports* (Springer Nature), 2025, 2026
- *Security Journal* (Springer Nature), 2025, 2026
- *Scientific Data* (Nature Publishing Group), 2025
- *Applied Network Science* (Springer Nature), 2025
- *European Transport Research Review* (Springer Nature), 2024
- *Applied Intelligence* (Springer Nature), 2022, 2023

Conference Reviewing

- Transportation Research Board (TRB) Annual Meeting, paper reviewer (2022, 2023, 2025)

Leadership & Service

- Coordinator, Graduate Writing Lab, College of Engineering, UMass Amherst (Spring 2025). Facilitated weekly writing sessions and provided feedback on graduate student manuscripts.
- Treasurer, Graduate Student Association, Dept. of Civil and Environmental Engineering, UMass Amherst (Sep 2021 – Sep 2022). Managed budgeting and event coordination for graduate student activities.

TECHNICAL SKILLS & METHODS

Statistical Modeling	Bayesian inference, mixture models, latent class analysis, ordered outcome models
Machine Learning	Gradient boosting, unsupervised clustering, dimensionality reduction, interpretable ML (SHAP)
Optimization	Mixed-integer linear programming, network optimization
Survey Methods	Stated preference and questionnaire design, incentivized panel deployment, response quality assurance and validation
Geospatial	Census-level spatial analysis, transit network analysis, GIS
Transportation	Transit demand modeling, OD inference, active mobility and micromobility, road safety and equity analysis

Software	Python, SQL, Git, GeoPandas, QGIS, Scikit-learn, XGBoost, Pandas

PROFESSIONAL AFFILIATIONS

- ELEVATE Affiliate, Energy Transition Institute, UMass Amherst
- Institute of Transportation Engineers (ITE)
- American Society of Civil Engineers (ASCE)
- Institute for Operations Research and the Management Sciences (INFORMS)
- Institute of Electrical and Electronics Engineers (IEEE)

REFERENCES

Dr. Jimi Oke	Associate Professor, Civil & Environmental Engineering, UMass Amherst (PhD advisor), jimi@umass.edu
Dr. Eleni Christofa	Professor, Civil & Environmental Engineering, UMass Amherst (committee member), echristofa@umass.edu
Dr. Camille Barchers	Associate Professor, Landscape Architecture & Regional Planning, UMass Amherst (committee member), cbarchers@umass.edu